

2024



Deluxe Media Inc. Processes over 150,000 Assets per Month Using AWS Batch

Learn how Deluxe Media Inc. in the media and entertainment industry scaled its batch processing capacity by five times using AWS Batch.

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5x increase

in vCPU scaling capacity

Hun

of job



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50% decrease

in error rate from 2–2.5% to 1%



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by thousands of dollars per month

Reduced monthly usage

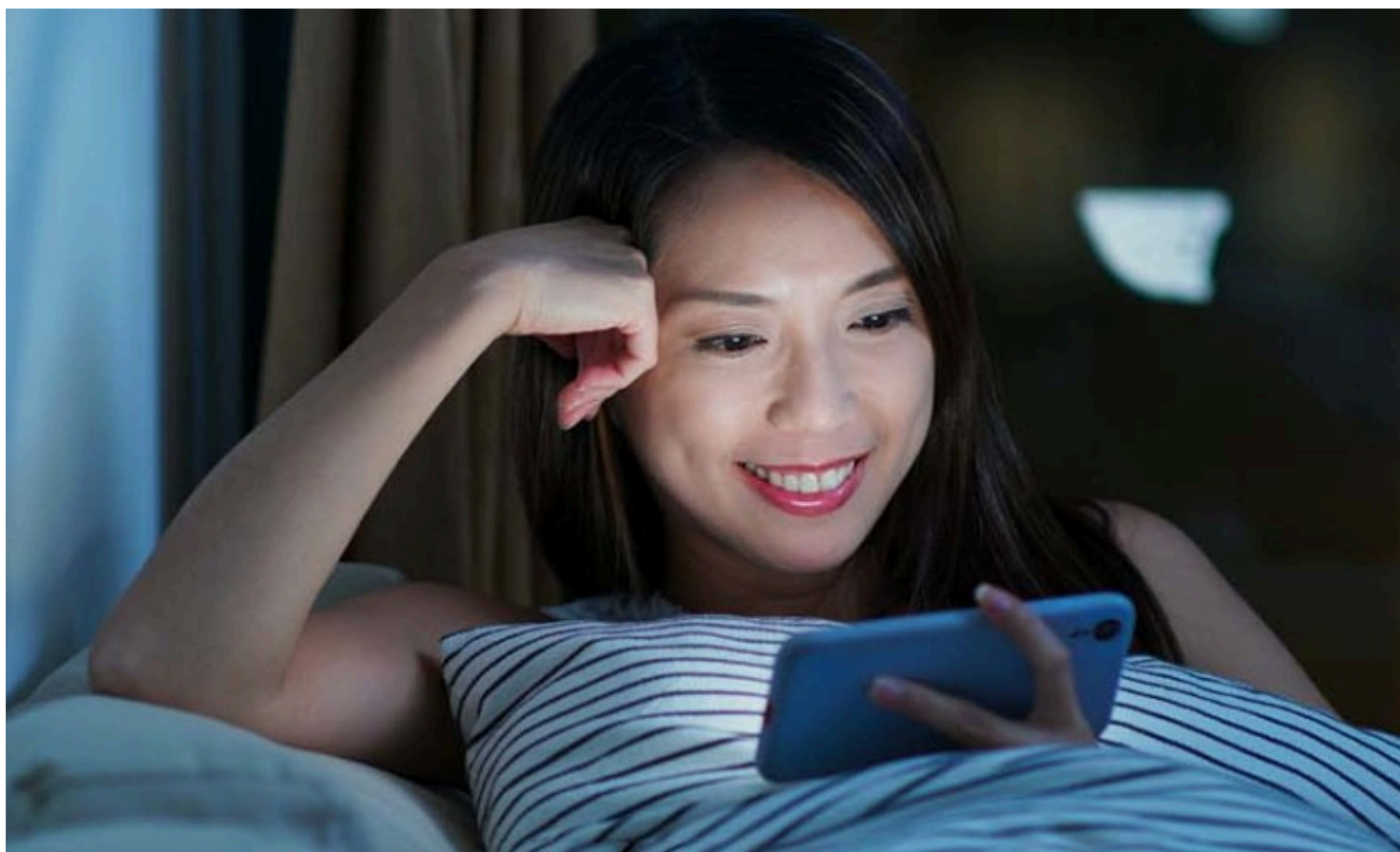


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Overview

Media services provider [Deluxe Media Inc.](#) (Deluxe) helps distribute content for some of the biggest names in the media industry. The company has been around since 1915 and has evolved to stay on the cutting edge of technology. Its cloud-native solution, Deluxe One, runs on Amazon Web Services (AWS) and processes over 150,000 assets per month.

To keep up with technological changes—such as delivering new streaming formats for new devices, including higher-resolution video—Deluxe needs a reliable, scalable, cloud-native solution for batch processing. It began using [AWS Batch](#), a fully managed batch computing service that plans, schedules, and runs containerized batch workloads. Since migrating to AWS Batch from another commercial self-managed batch solution, Deluxe has improved its scalability, performance, and cost efficiency.



Opportunity | Using AWS Batch to Orchestrate Thousands of Batch Jobs per Month for Deluxe

Deluxe transforms, localizes, and distributes content for some of the world's largest content creators, broadcasters, and streaming platforms. Its digital media supply chain service, Deluxe One, is built on AWS and composed of over 500 containerized microservices running on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. These microservices perform functions such as transcoding, compression, applying digital rights management (DRM) encryption, and running checksums.



Previously, Deluxe used a self-managed batch orchestrator to manage the hundreds of thousands of batch jobs that these microservices generate per month, but this solution created a bottleneck. If Deluxe scaled its solution too quickly, the batch orchestrator experienced stability issues. “Scalability is incredibly important,” says Jeff Cuneo, senior director of platform engineering at Deluxe Media. “On AWS, we knew we weren’t going to run out of Amazon EC2 instances or storage space, but we couldn’t consume them as quickly or at the scale that we wanted to because of the limitations of the previous batch provider.”

To mitigate the stability issues with the old solution, Deluxe capped the number of virtual CPUs (vCPUs) that its batch provider could run in each compute pool. Deluxe engineers knew that they needed to find a better long-term solution, especially as Deluxe continued to grow and process larger assets for customers who had tight delivery schedules and exacting service-level agreements in place. Deluxe began talking with the AWS team about using AWS Batch. The company wanted a cloud-native solution, and Deluxe worked with the AWS team to thoroughly load-test AWS Batch before migrating its containers to the service—with virtually no downtime. Unlike the previous solution, AWS Batch runs hundreds of thousands of batch computing jobs without installing software or servers. Deluxe began testing AWS Batch in February 2023 and went live using the service in July 2023.

Solution | Scaling Job Capacity by Five Times while Cutting Costs

In AWS Batch, Deluxe creates unique job queues and compute environments optimized for the type of job that is going to run through it. Deluxe One runs many different types of jobs with various parameters. For example, the solution applies DRM encryption so that files can be read only by those with the right authorization or in the correct region. Because DRM encryption requires many more input/output actions, those DRM jobs are placed in a compute pool using Amazon EC2 instances optimized for input/output operations.

Deluxe One is more scalable now that it runs using AWS Batch. “Using AWS Batch, we’re able to scale up to higher levels much faster and get more work done quickly,” says Cuneo. Deluxe previously set an artificial limit of 400 vCPUs per compute pool to avoid stability issues. Now, by running on AWS Batch, Deluxe could move the cap to 2000 vCPUs, which is more than it needs for its daily use, and can go to a much higher number of vCPUs if needed.

After the change, the Deluxe One pipeline reduced its error rate by 50 percent, from a 2–2.5 percent error rate to just 1 percent. With the solution running 600,000 to 800,000 batch jobs per month, that diminished the number of jobs that fail, which streamlines the operations teams’ work because they have fewer jobs to remediate. “Every failed job is a yes



from an employee to mitigate it, even if the remediation is as simple as hitting a restart button, so this reduction accounts for hundreds of hours per month,” says Cuneo.

By adopting AWS Batch, Deluxe also removed artificial limits on how quickly its batch queue can scale, meaning that it can respond to an influx of jobs more quickly. With the ability to perform more simultaneous jobs, Deluxe can provide more reliable service to customers and better meet its service-level agreements. Deluxe also reduced compute costs by thousands of dollars each month. Previously, the service would often run nodes up to 3 hours longer than necessary as it slowly scaled down from its maximum capability to baseline. Now, it can scale from 2000 vCPUs to zero in 5 minutes if needed, which reduces unnecessary compute spending and cuts down the monthly usage for its live environments from 150,000 GB to 90,000 GB. Deluxe further optimized its costs in [Amazon Elastic Block Store](#) (Amazon EBS), a simple-to-use, scalable, high-performance block-storage service for Amazon EC2. It also removed the licensing costs associated with its previous batch orchestrator, and it benefits from using [Amazon EC2 Spot Instances](#)—which are available at up to a 90 percent discount compared with Amazon EC2 On-Demand Pricing.

Deluxe can now confidently take on new workloads without worrying about the ability to deliver on time and meet service-level agreements. “We’re ready for new clients anytime,” says Cuneo. “Using AWS Batch, we know that we can handle the volume.” In addition to AWS Batch, Deluxe One uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage solution built to retrieve virtually any amount of data from anywhere. “Storage is incredibly important for this type of media workflow, and on premises it can be the ultimate bottleneck,” says Cuneo. “Using Amazon S3, we’re not constrained by how much storage we can access or how fast the storage is. Amazon S3 serves as one of the linchpin AWS services for our solution.”

Outcome | Expanding Capacity for Growth on AWS

The migration to AWS Batch has increased Deluxe’s capacity to scale and handle more work, not only in volume but also in its ability to process larger assets. Using AWS Batch, Deluxe can confidently take on new workloads without worrying about its ability to deliver on time and meet its service-level agreements.

Migrating its batch service is the first part of a larger improvement plan for Deluxe’s environment. “We are continually driving additional workloads to the cloud in large part because the support that the AWS team gives us makes it simple to migrate,” says Cuneo.

About Deluxe Media Inc.

Deluxe Media Inc. provides services to transform, localize, and distribute content for some of the largest content creators, broadcasters, and streaming platforms in the world. Its digital media supply chain service, Deluxe One, is built on AWS.

AWS Services Used



Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon EBS

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Batch

AWS Batch is a fully managed batch computing service that plans, schedules, and runs your containerized batch ML, simulation, and analytics workloads across the full range of AWS compute offerings, such as Amazon ECS, Amazon EKS, AWS Fargate, and Spot or On-Demand Instances.

[Learn more »](#)

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ABP Network Uses Generative AI to Produce...

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Ringier Axel Springer Polska Reduces Carbon...

Ringier Axel Springer Polska (RASP) drastically reduced its carbon emissions after migrating to AWS. RASP, a Polish digital media company, committed to reducing carbon emissions by 90 percent. As part of a strategy geared toward compute efficiency and price performance, it migrated to Amazon EC2 Graviton processors, which cut its carbon emissions for IT systems to near



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